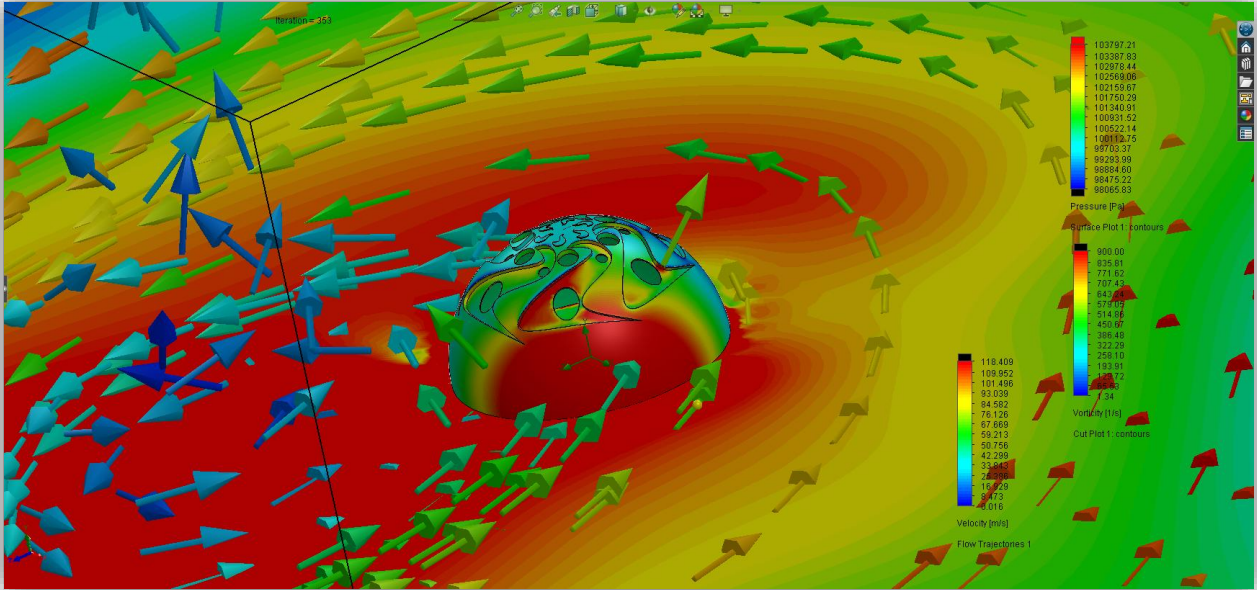




WENDIBALL

Computational Fluid Dynamics Study

Preliminary Design Report — Iterations 1 through 3

Wendell Abara

May 2026

SOLIDWORKS Flow Simulation • Steady-State CFD • ABS Plastic • 72 mm Diameter

1. Project Overview

The Wendiball project is an original aerodynamic ball design created by Wendell Abara. The central concept is that surface geometry — specifically vane patterns and through-hole arrangements — can be used to engineer asymmetric aerodynamic forces during flight, enabling a predictable and controllable trajectory break when the ball is thrown with spin.

This report documents the first phase of the project: a computational fluid dynamics (CFD) study conducted in SOLIDWORKS Flow Simulation across three design iterations and a smooth-ball baseline. The simulations are intended to serve as a comparative aerodynamic tool to guide physical prototype selection. No physical testing has been conducted at the time of writing; that work is planned as a subsequent phase.

1.1 Motivation and Inspiration

The Wendiball project originated from a long-standing interest in the aerodynamics of breaking pitches in baseball. Pitch types such as the curveball, sweeper, and slider each produce a distinct trajectory break through a combination of spin axis orientation, spin rate, and velocity. What makes these pitches so difficult to hit is precisely what makes them interesting from an engineering standpoint: the break is not random. It is a predictable consequence of aerodynamic forces acting on a rotating sphere in flight.

The question that eventually launched this project was: why do wiffleball pitches break so dramatically? A wiffleball thrown with the same spin as a baseball produces trajectory deviations that appear almost cartoonishly exaggerated by comparison. The same observation applies to blitzball, another perforated recreational ball with a reputation for extreme, sharply breaking pitches. The answer lies in the ball's geometry. The asymmetric hole pattern on a wiffleball disrupts flow symmetry in a way that a smooth sphere cannot, generating lateral aerodynamic forces far beyond what the Magnus effect alone would produce at the same velocity and spin rate. The holes are not incidental — they are the mechanism.

This observation raised a follow-on question: what would happen if that principle were pushed further? What if the surface geometry of the ball were designed not incidentally — the wiffleball holes exist primarily for lightweight construction — but intentionally, with the specific goal of engineering a larger, more directional, or more controllable aerodynamic break? The idea was effectively to design a “wiffleball on steroids”: a ball that retains the basic form factor and throwing mechanics of a recreational projectile, but uses surface vane patterns, hole arrangements, or both, to generate aerodynamic forces well beyond what a smooth sphere or a randomly-holed ball could produce.

To investigate this, SOLIDWORKS Flow Simulation was used to run steady-state CFD studies on custom-designed ball geometries. Alongside the simulations, analytical calculations using Reynolds number theory and aerodynamic force modeling helped characterize the expected flow regime and validate the simulation setup. The design space was deliberately constrained to match realistic backyard throwing conditions — the same velocity and spin ranges a human thrower might produce — so that the results would be meaningful in a practical recreational context, not just a laboratory one. Out of that process came three design iterations and this report.

2. Global Simulation Parameters

All simulations in this study share the following fixed parameters, ensuring that aerodynamic differences between iterations are attributable to geometry alone.

2.1 Ball Properties (All Iterations)

Parameter	Value
Diameter	72 mm
Wall Thickness	1.2 mm
Material	ABS Plastic
Simulation Type	Steady-State CFD
Software	SOLIDWORKS Flow Simulation

Table 1. Fixed ball properties shared across all iterations.

2.2 Simulation Conditions

Simulation	Scenario	Velocity	RPM
Sim 1	Backyard Fastball	30 m/s (70 mph)	2000 RPM
Sim 2	Regular Backyard Throw	17.88 m/s (40 mph)	500 RPM

Table 2. Simulation conditions applied to every iteration.

The ball travels in the positive x-axis direction and spins clockwise about the y-axis in all simulations. Local and global mesh parameters, global rotation, and initial and ambient conditions are held constant across all runs.

3. Assumptions and Limitations

3.1 Modeling Assumptions

- Steady-state CFD is used throughout. Transient aerodynamic effects, including vortex shedding dynamics over time, are not captured.
- Material properties are idealized. ABS plastic is modeled with standard library values; surface roughness is not applied.
- Spin rates and throwing velocities are estimated based on realistic backyard throwing scenarios, not measured experimentally.
- Environmental conditions are simplified: standard sea-level air properties, no wind, no humidity effects.

3.2 Convergence Limitations

Convergence behavior varied significantly across simulations and represents the most important limitation on the quantitative interpretation of results. Goal convergence percentages ranged from approximately 14% to 100% across the various force and torque goals.

While goals with convergence near or above 50% can be used with reasonable confidence for comparative analysis, goals below 30% should be treated as indicative rather than definitive. The steady-state solver can struggle to fully converge in regions of strong vortex shedding and separated flow, which are present in all vanned iterations — particularly Iteration 1 at fastball conditions. Quantitative force comparisons are therefore treated as directional rather than precisely validated.

These limitations are expected and acceptable for the preliminary purpose of this study: guiding geometry selection ahead of physical prototype testing.

3.3 Other Limitations

- Mesh resolution around small vane features and localized vortex structures is constrained by computational resources.
- No physical experimental validation has been conducted. All results are preliminary engineering observations.
- CFD comparisons do not account for mass differences between designs. Physical testing will incorporate mass-normalized performance evaluation.

4. Iteration 1 — Spiral Vane with Through-Holes

4.1 Geometry Description

Iteration 1 features a spiral vane design combined with a pattern of through-holes distributed across the ball surface. The vanes and holes interact to create complex localized flow structures, separating and re-energizing boundary layer flow in an asymmetric pattern. Mass: 17.41 g.

Property	Value
Mass	17.41 g
Volume	17,071.24 mm ³
Surface Area	31,623.13 mm ²
Center of Mass (X, Y, Z)	(0.00, -3.19, 0.00) mm
Principal Moments of Inertia (Px, Py, Pz)	13,954 / 13,955 / 15,407 g·mm ²

Table 3. Mass properties of Iteration 1.

4.2 Simulation 1 — Fastball Conditions (30 m/s, 2000 RPM)

At fastball conditions, Iteration 1 produced strong and pronounced aerodynamic forces. The pressure surface plots show a highly asymmetric distribution, with a large high-pressure region on the windward hemisphere and a distinct low-pressure wake concentrated across the vane and hole structures. The flow trajectory plots reveal a large coherent spiral wake with peak velocities reaching approximately 148 m/s near the vanes.

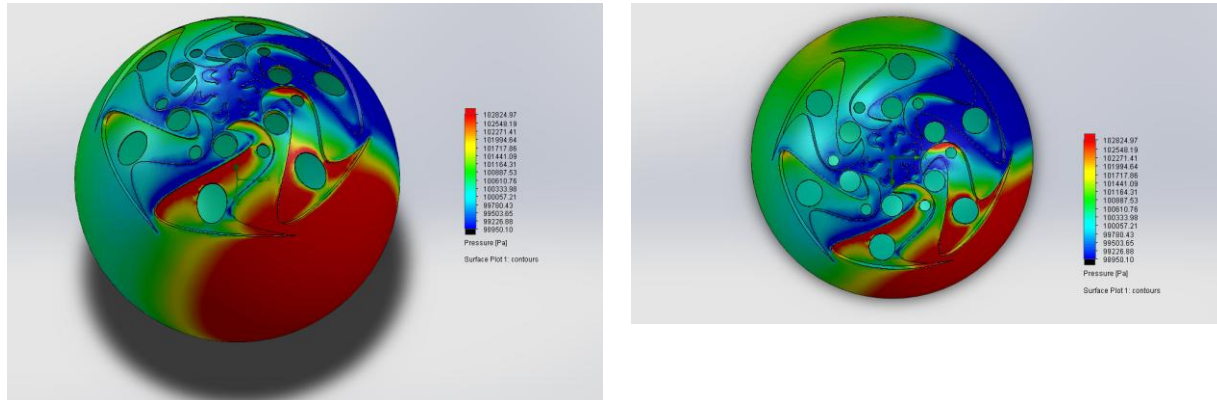


Figure 1. Iteration 1 Sim 1 — Pressure surface plot, front view (left) and top view (right). Strong asymmetric pressure distribution characteristic of the vane-and-hole geometry.

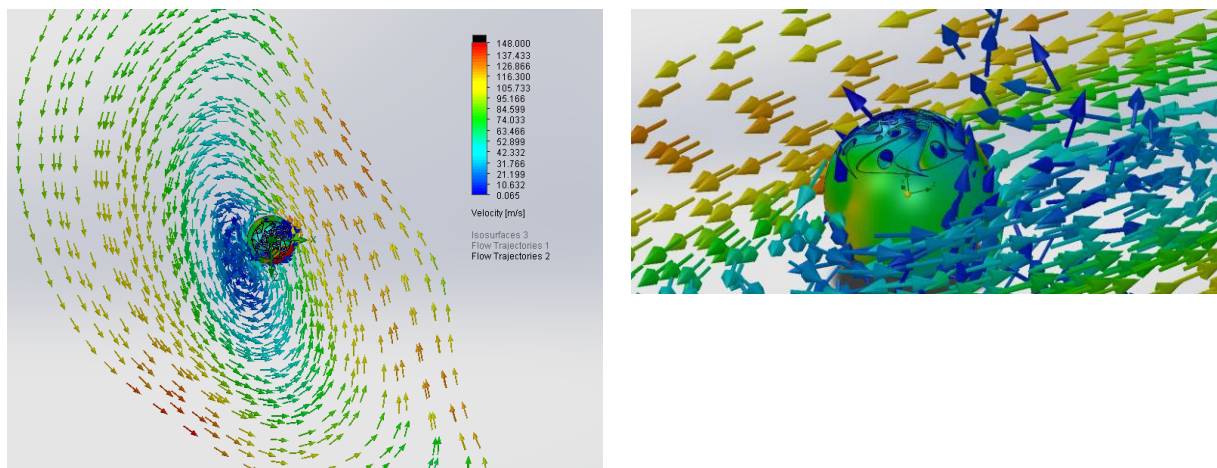


Figure 2. Iteration 1 Sim 1 — Flow trajectory (velocity) with pressure surface overlay. Left: zoomed-out top-plane view showing large-scale spiral wake. Right: close-up of the vane region.

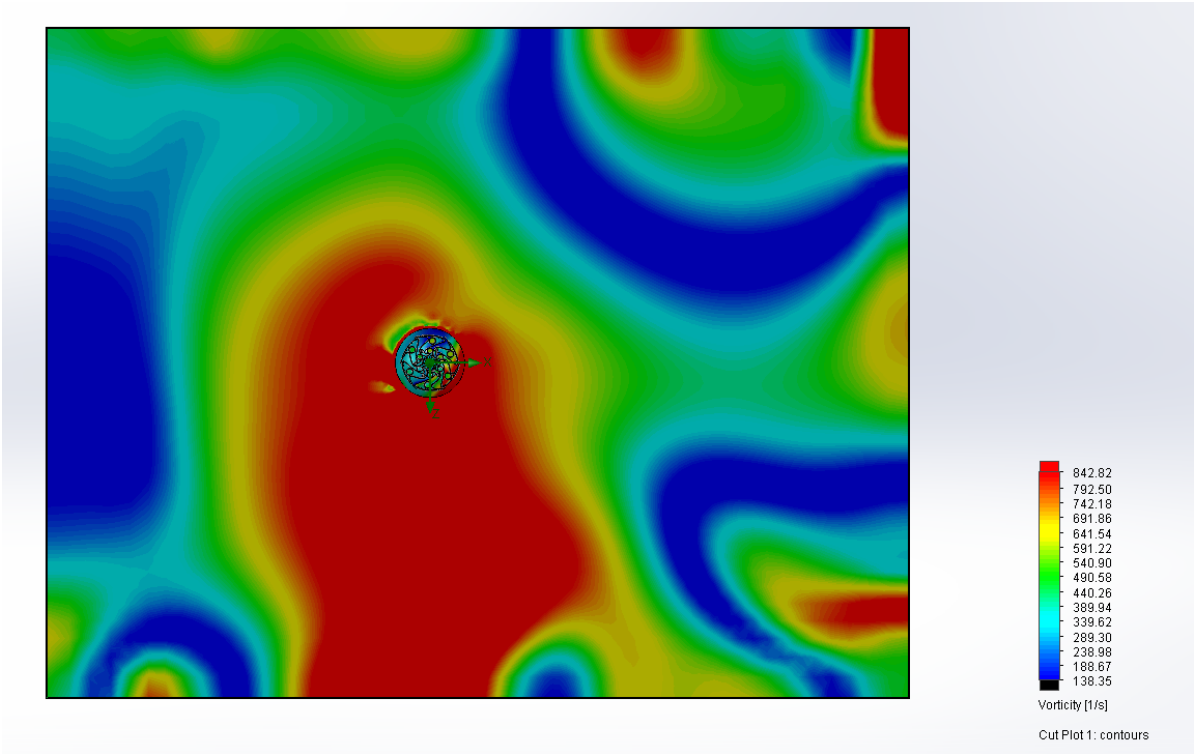


Figure 3. Iteration 1 Sim 1 — Vorticity cut plot through the equatorial plane. High-rotation zones extend well beyond the ball diameter.

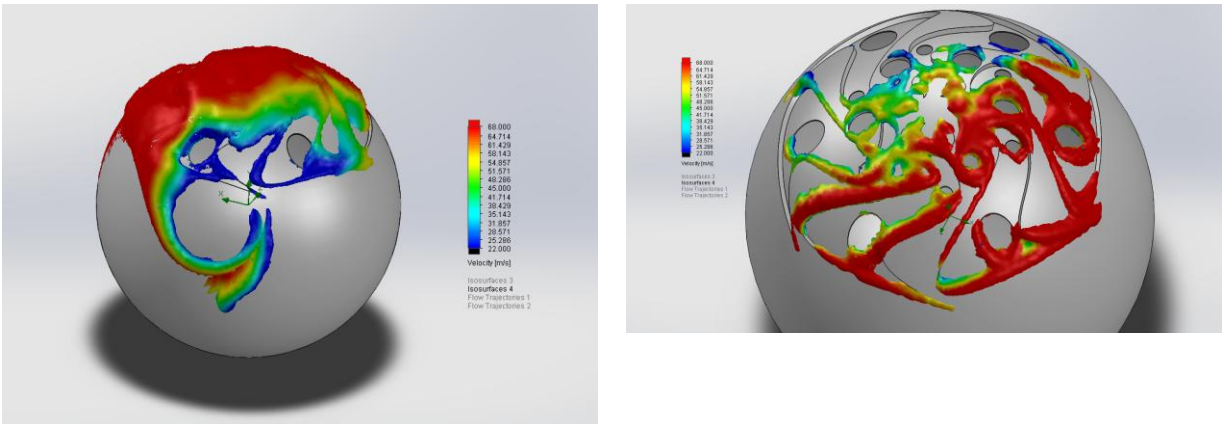


Figure 4. Iteration 1 Sim 1 — Vorticity isosurfaces coloured by velocity. Left: threshold at 10,000 s⁻¹. Right: threshold at 30,000 s⁻¹, revealing intense vortex structures concentrated along vane leading edges.

Metric	Value	Notes
Force X (Axial)	-2.32 N	Drag-direction force
Force Y (Lateral)	-9.13 N	Strong side force from vane asymmetry
Force Z (Directional)	-11.30 N	Dominant directional force
Normal Force Y	-9.12 N	Confirms strong asymmetric surface loading
Torque Y	0.00124 N·m	Aerodynamic spin-axis torque

Resultant XYZ Force	~14.71 N	Total aerodynamic force magnitude
Solver Iterations	647	
Convergence	~17–100%	Variable; Y-force goals partially converged

Table 4. Iteration 1 Sim 1 (Fastball) — CFD goal summary.

4.3 Simulation 2 — Regular Throw Conditions (17.88 m/s, 500 RPM)

At regular throw conditions, aerodynamic forces drop substantially. The pressure distribution becomes softer and more localized. The wake structure is notably smaller and more fragmented, consisting of weaker local vortex structures rather than the coherent high-energy wake seen in Sim 1.

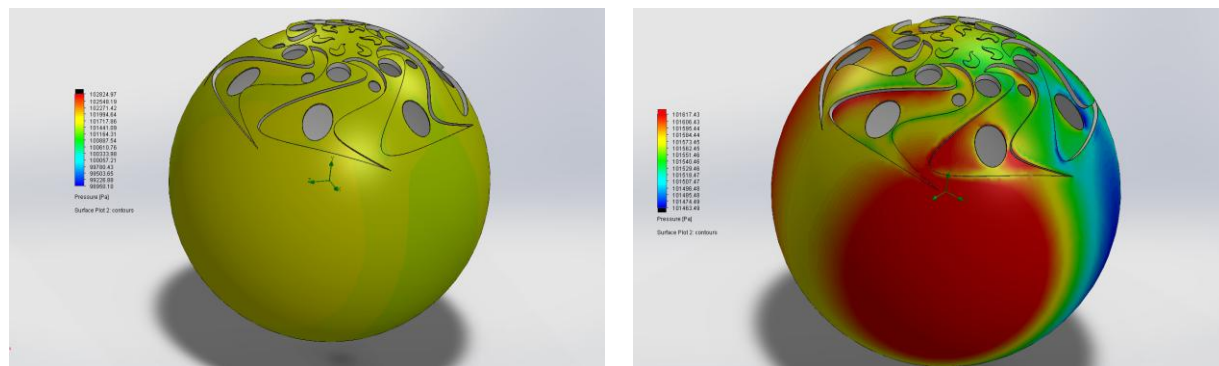


Figure 5. Iteration 1 Sim 2 — Pressure surface plots. Left: same scale as Sim 1, showing the dramatically reduced pressure differential at 40 mph. Right: scale adjusted to the 40 mph case, revealing subtle localized gradients around vane edges.

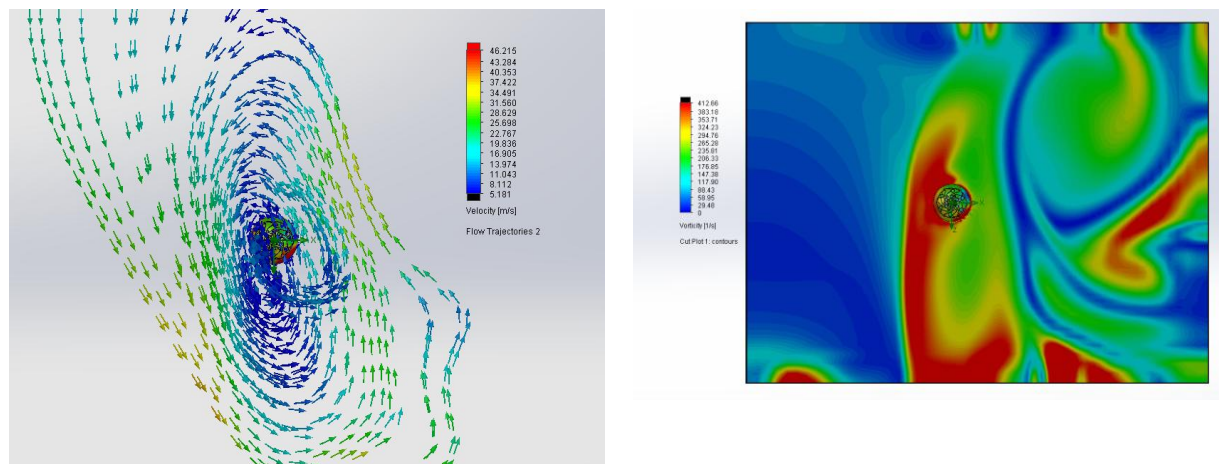


Figure 6. Iteration 1 Sim 2 — Left: flow trajectory top view, showing a much smaller and less energetic spiral wake. Right: equatorial vorticity cut plot on the same scale as Sim 1, confirming the far weaker rotational field.

Metric	Value	Notes
Force X	0.070 N	Near-zero; slight sign change vs. Sim 1
Force Y (Lateral)	-0.106 N	~86× smaller than Sim 1

Force Z (Directional)	-0.152 N	~74× smaller than Sim 1
Normal Force Y	-0.106 N	Greatly reduced surface asymmetry
Torque Y	5.63×10^{-5} N·m	~22× lower than Sim 1
Resultant XYZ Force	~0.198 N	~74× smaller than Sim 1
Solver Iterations	699	
Convergence	~14–26%	Lower convergence throughout

Table 5. Iteration 1 Sim 2 (Regular Throw) — CFD goal summary.

4.4 Iteration 1 Summary

Iteration 1 demonstrates strong sensitivity to throwing speed and spin rate. At fastball conditions the vane-and-hole geometry generates substantial asymmetric forces, particularly in the lateral Y-direction. At regular throw conditions the same geometry produces only subtle aerodynamic disturbances. The fragmented turbulence-heavy wake, while producing significant lateral force, suggests less predictable trajectory behavior compared to a more coherent flow-guiding design.

5. Iteration 2 — High-Count Thin Spiral Vanes

5.1 Geometry Description

Iteration 2 removes the through-holes from Iteration 1 and replaces the vane geometry with a higher-count pattern of thinner spiral vanes. The intent is to isolate the vane contribution to aerodynamic performance and explore whether denser, thinner vanes can generate more coherent directional flow guidance.

Mass properties of John 2 Configuration: Default Coordinate system: -- default --		
Density = 0.00 grams per cubic millimeter		
Mass = 18.76 grams		
Volume = 18392.15 cubic millimeters		
Surface area = 32000.23 square millimeters		
Center of mass: (millimeters)		
X = 0.00		
Y = -0.77		
Z = 0.00		
Principal axes of inertia and principal moments of inertia: (grams * square m		
Taken at the center of mass.		
Ix = (0.91, 0.00, -0.41)	Px = 15567.98	
Iy = (-0.41, 0.00, -0.91)	Py = 15569.12	
Iz = (0.00, 1.00, 0.00)	Pz = 15855.62	
Moments of inertia: (grams * square millimeters)		
Taken at the center of mass and aligned with the output coordinate system.		
Lxx = 15568.17	Lxy = 0.90	Lxz = -0.43
Lyx = 0.90	Lyj = 15855.62	Lyz = 0.81
Lzx = -0.43	Lzy = 0.81	Lzz = 15568.93
Moments of inertia: (grams * square millimeters)		
Taken at the output coordinate system. (Using positive tensor notation.)		
Ixx = 15579.21	Ixy = 0.88	Ixz = -0.43
Iyx = 0.88	Iyy = 15855.62	Iyz = 0.79
Izx = -0.43	Izy = 0.79	Izz = 15579.97

Figure 7. Iteration 2 geometry — high-count thin spiral vane pattern, no through-holes.

Property	Value
Mass	18.76 g
Volume	18,392.15 mm ³
Surface Area	32,000.23 mm ²
Center of Mass (X, Y, Z)	(0.00, -0.77, 0.00) mm
Principal Moments of Inertia (Px, Py, Pz)	15,568 / 15,569 / 15,856 g·mm ²

Table 6. Mass properties of Iteration 2. Mass is ~1.35 g greater than Iteration 1 due to absence of through-holes.

5.2 Simulation 1 — Fastball Conditions (30 m/s, 2000 RPM)

Iteration 2 produces a notably smoother and more coherent pressure distribution than Iteration 1. The thinner, higher-count vanes guide airflow more directionally, functioning more like an impeller than a turbulence generator. Peak velocities near the vane surfaces reach approximately 165 m/s — higher than Iteration 1 at the same conditions. Isosurface vorticity plots show strong vortex filaments aligned with individual vane trailing edges, indicating organized rather than chaotic vortex shedding.

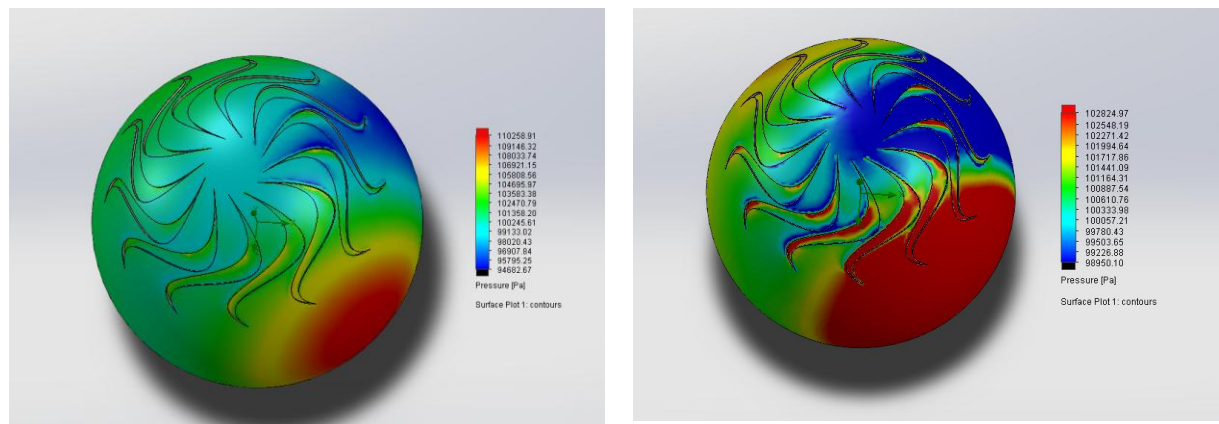


Figure 8. Iteration 2 Sim 1 — Pressure surface plots. Left: on the same scale as Iteration 1 Sim 1. Right: pressure with flow trajectory overlay, highlighting directional flow channels along the vane geometry.

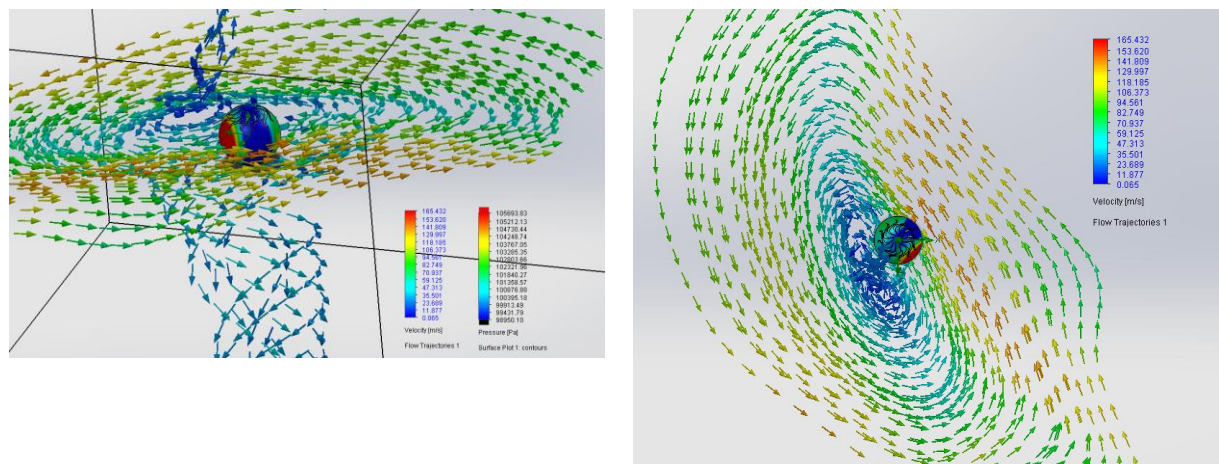


Figure 9. Iteration 2 Sim 1 — Flow trajectory (velocity). Left: 3D isometric view showing the organized spiral wake structure. Right: top-plane view showing the well-defined rotational flow field.

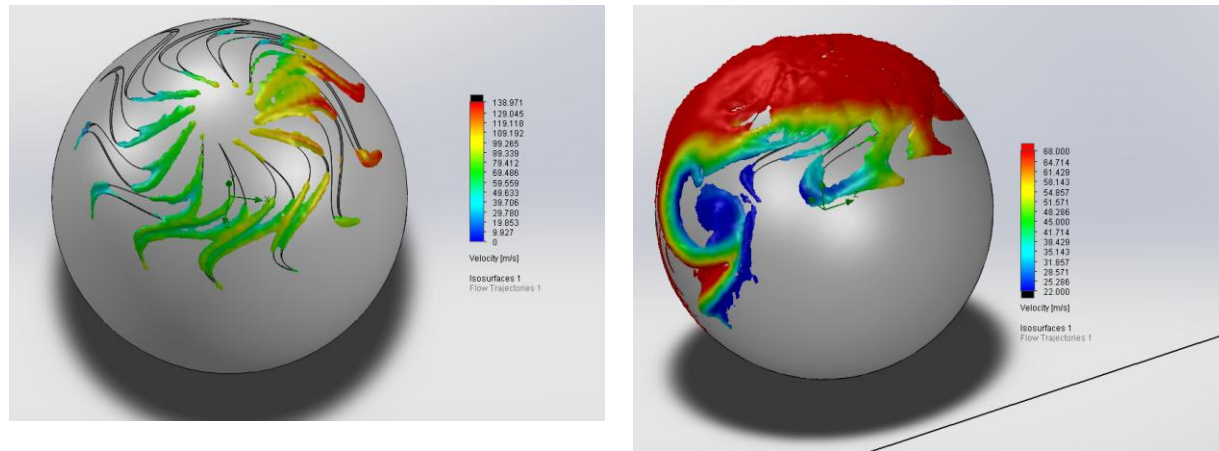


Figure 10. Iteration 2 Sim 1 — Vorticity isosurfaces. Left: threshold at $50,000 \text{ s}^{-1}$ (scale adjusted), revealing individual vane-trailing vortex filaments. Right: threshold at $10,000 \text{ s}^{-1}$ on the same scale as Iteration 1, confirming a more concentrated and coherent vortex field.

Metric	It. 2 Sim 1	vs. It. 1 Sim 1	Notes
Force X	-5.03 N	+2.71 N larger	Stronger axial drag
Force Y (Lateral)	-5.28 N	-3.85 N smaller	Less lateral disturbance
Force Z (Directional)	-15.21 N	+3.91 N larger	Much stronger directional force
Normal Force Y	-5.27 N	Smaller asymmetry	More symmetric surface loading
Torque Y	0.00166 N·m	+34% higher	Higher rotational torque
Resultant Force	~16.89 N	+15% larger	Overall strongest vaned design
Convergence	~30–95%	Better	Healthier convergence

Table 7. Iteration 2 Sim 1 (Fastball) — comparison with Iteration 1 Sim 1.

5.3 Simulation 2 — Regular Throw Conditions (17.88 m/s, 500 RPM)

At regular throw conditions, Iteration 2 continues to outperform Iteration 1 in overall force magnitude. The pressure surface plots show the smoothest gradient of any vaned design at slow speed, and convergence is notably better.

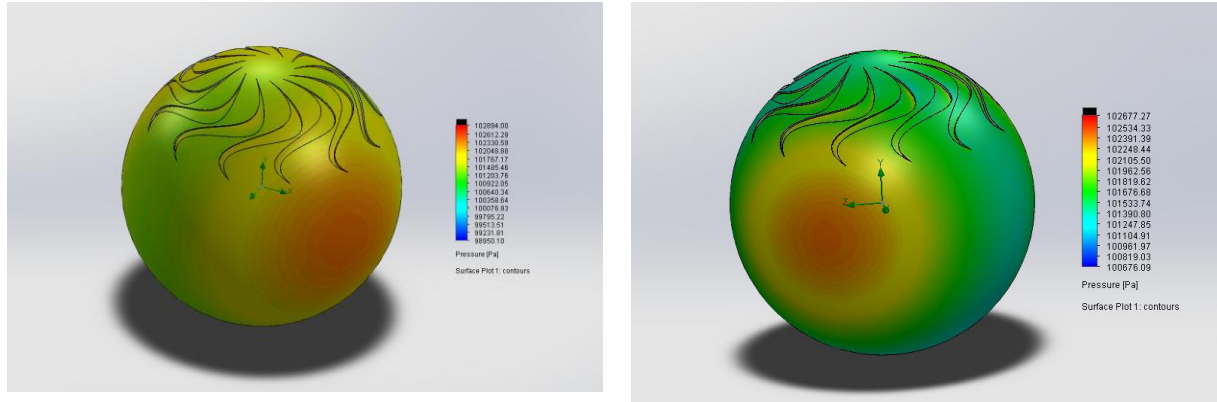


Figure 11. Iteration 2 Sim 2 — Pressure surface plots. Left: same scale as Iteration 1 Sim 1 reference. Right: scale adjusted to the 40 mph case, showing a smooth and well-distributed pressure gradient across the vane surfaces.

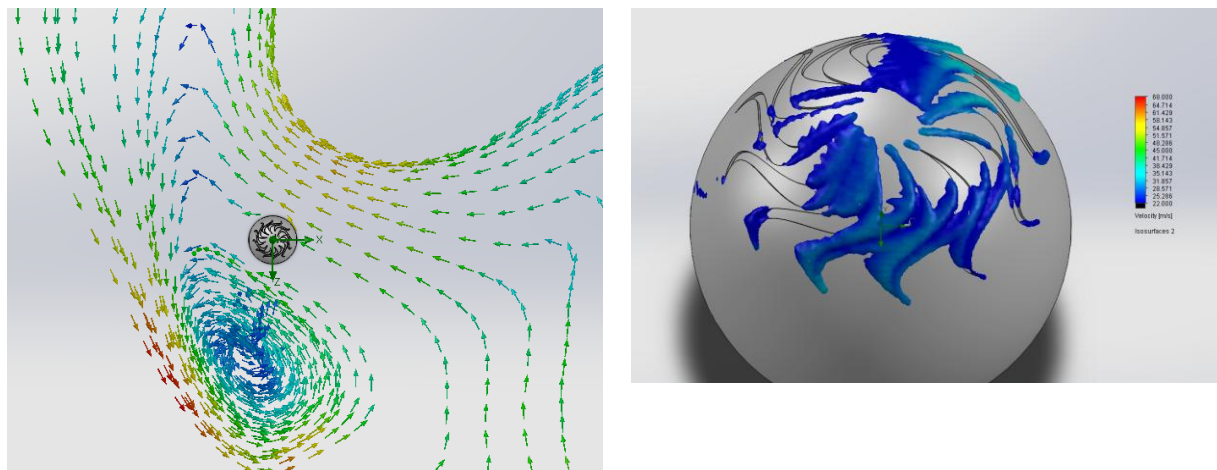


Figure 12. Iteration 2 Sim 2 — Left: flow trajectory top view showing a coherent rotational wake. Right: vorticity isosurface at $10,000 \text{ s}^{-1}$ on the same scale as Iteration 1, showing concentrated and organized vortex structures along individual vanes.

Metric	It. 2 Sim 2	vs. It. 1 Sim 2	Notes
Force X	-0.279 N	~4.0× larger	Much stronger axial force
Force Y (Lateral)	-0.283 N	~2.7× larger	Significantly more lateral force
Force Z	-0.275 N	~1.8× larger	Stronger directional force
Torque Y	$2.82 \times 10^{-5} \text{ N} \cdot \text{m}$	~50% lower	Reduced spin-axis torque
Resultant Force	~0.483 N	~2.4× larger	Dominant at slow throw
Convergence	~30–83%	Much better	Most stable regular-throw sim

Table 8. Iteration 2 Sim 2 (Regular Throw) — comparison with Iteration 1 Sim 2.

5.4 Iteration 2 Summary

Iteration 2 demonstrates that removing through-holes and increasing vane density produces a more aerodynamically coherent and directionally effective design. It generates the highest resultant force

of all vaned configurations tested, with the dominant contribution shifting toward the Z (directional) axis. The trade-off is a reduction in lateral Y-force relative to Iteration 1, which may translate to a different break character in physical flight.

6. Iteration 3 — Smooth Ball Baseline

6.1 Purpose

Iteration 3 is a perfectly smooth spherical shell with the same diameter and wall thickness. It establishes the aerodynamic baseline — the forces produced purely by the Magnus effect and standard sphere drag, with no surface geometry modification.

Property	Value
Mass	19.28 g
Volume	18,899.02 mm ³
Surface Area	31,504.39 mm ²
Center of Mass (X, Y, Z)	(0.00, 0.00, 0.00) mm
Moments of Inertia (Ixx = Iyy = Izz)	16,112 g·mm ² (all equal — perfect symmetry)

Table 9. Mass properties of Iteration 3 (smooth sphere baseline).

6.2 Simulation 1 — Fastball Conditions

The smooth ball produces a clean symmetric pressure distribution with a well-defined stagnation region on the windward face. The flow trajectory visualizations show a tightly wound helical wake, characteristic of a spinning sphere with a strong Magnus-type response. There are no localized vortex structures — the entire aerodynamic response originates from the interaction of translational and rotational flow around a symmetric body.

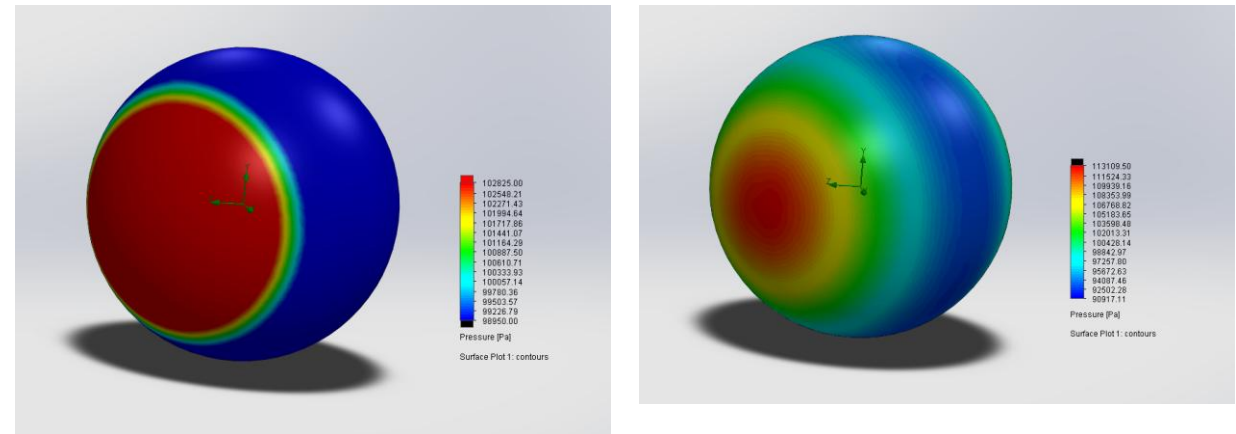


Figure 13. Smooth ball Sim 1 — Pressure surface plots. Left: on the same scale as Iteration 1, showing the smooth and symmetric pressure gradient. Right: scale adjusted to reveal the full pressure range, confirming a clean stagnation zone and broad low-pressure wake.

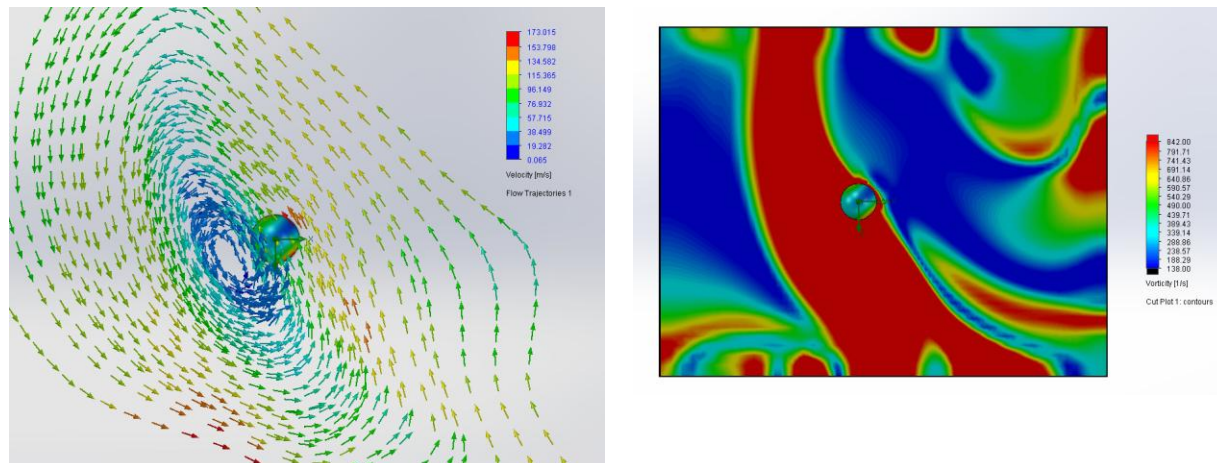


Figure 14. Smooth ball Sim 1 — Left: flow trajectory top view showing the clean helical Magnus wake. Right: equatorial vorticity cut plot, confirming the absence of localized turbulent structures.

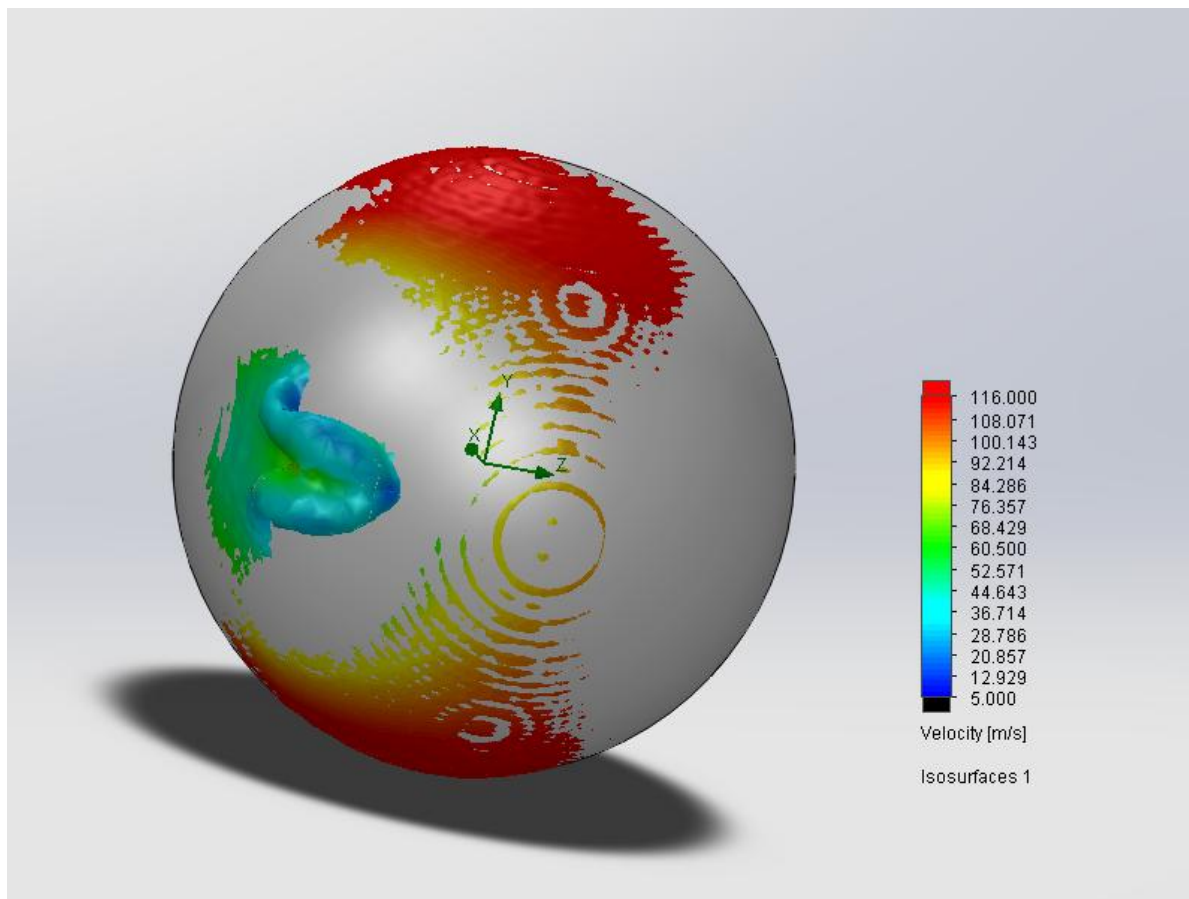


Figure 15. Smooth ball Sim 1 — Vorticity isosurface at $10,000 \text{ s}^{-1}$. The minimal and diffuse vortex structure confirms the smooth ball produces no localized vortex shedding.

Metric	Smooth Ball Sim 1	Notes
Force X (Axial)	-5.08 N	Standard sphere drag
Force Y (Lateral)	-0.21 N	Very low lateral force
Force Z (Directional)	-15.87 N	Strong Magnus-type Z-force
Normal Force Y	-0.20 N	Near-symmetric surface loading
Torque Y	0.00178 N·m	Highest torque of all designs
Resultant Force	~16.66 N	High total force from Magnus effect

Table 10. Smooth ball Sim 1 (Fastball) — CFD goal summary.

6.3 Simulation 2 — Regular Throw Conditions

At regular throw conditions, the smooth ball continues to produce a clean Magnus-dominated response. The pressure field is smooth and symmetric, and the wake is a tightly organized helical structure.

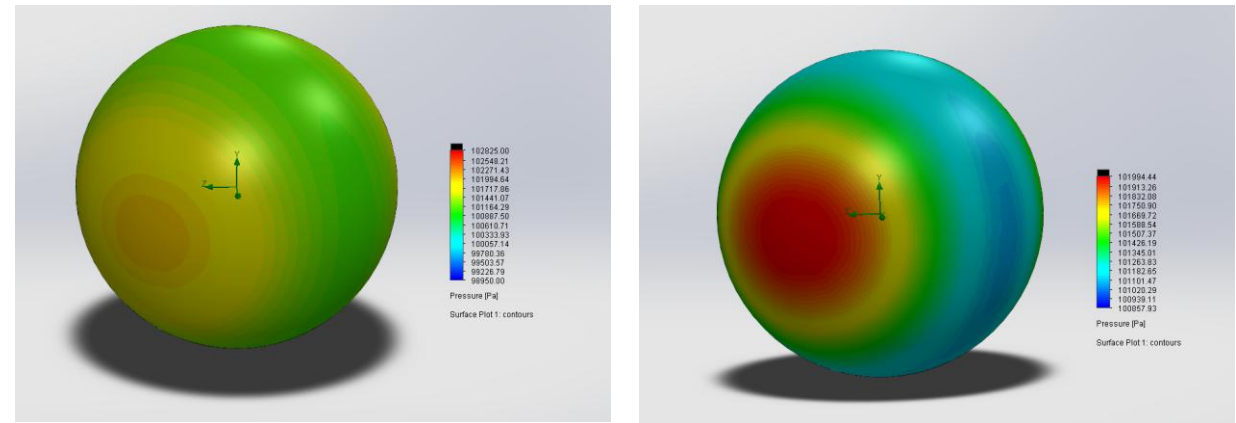


Figure 16. Smooth ball Sim 2 — Left: pressure surface plot (adjusted scale), showing the smooth gradient characteristic of a symmetric sphere. Right: flow trajectory side view, showing the clean helical wake at 40 mph / 500 RPM.

Metric	Smooth Ball Sim 2	Notes
Force X	-0.320 N	
Force Y	-0.0159 N	Near-zero lateral force
Force Z	-0.325 N	Dominant directional component
Torque Y	3.95×10 ⁻⁶ N·m	
Resultant Force	~0.456 N	

Table 11. Smooth ball Sim 2 (Regular Throw) — CFD goal summary.

6.4 Smooth Ball Summary

The smooth ball establishes that spin-induced aerodynamic force is already substantial before any surface geometry is added. The vanned designs are best understood as modifiers of this baseline Magnus response rather than generators of force from scratch. The key question is how each design

redistributes force between the Z and Y axes, and whether the modification is predictable and controllable.

7. Cross-Iteration Comparison

7.1 Fastball Conditions (30 m/s, 2000 RPM)

Metric	Smooth Ball	Iteration 1	Iteration 2	Interpretation
Geometry	Smooth sphere	Vanes + holes	Thin vanes, no holes	Three aerodynamic control levels
Force X	-5.08 N	-2.32 N	-5.03 N	It. 2 matches smooth ball drag
Force Y (Lateral)	-0.21 N	-9.13 N	-5.28 N	Vanes introduce major lateral force
Force Z (Directional)	-15.87 N	-11.30 N	-15.21 N	It. 2 preserves strong Z-force
Torque Y	0.00178 N·m	0.00124 N·m	0.00166 N·m	It. 2 closest to smooth ball torque
Resultant Force	~16.66 N	~14.71 N	~16.89 N	It. 2 highest overall
Wake Character	Clean helical	Chaotic turbulent	Organized directional	Distinct behaviors
Pressure Field	Smooth radial	Fragmented/localized	Smooth coherent gradient	It. 2 guides flow cleanly

Table 12. Fastball condition comparison across all three iterations.

Iteration 1 disrupts the smooth-ball Magnus response and converts a large portion of Z-force into lateral Y-force through its turbulent, chaotic wake. Iteration 2 preserves most of the smooth ball's Z-force while adding meaningful lateral force — a qualitatively different and more controlled aerodynamic profile. Iteration 2 also produces the highest total resultant force.

7.2 Regular Throw Conditions (17.88 m/s, 500 RPM)

Metric	Smooth Ball	Iteration 1	Iteration 2	Interpretation
Force X	-0.320 N	0.070 N	-0.279 N	Smooth and It. 2 similar axial response
Force Y (Lateral)	-0.0159 N	-0.106 N	-0.283 N	It. 2 creates strongest lateral force
Force Z	-0.325 N	-0.152 N	-0.275 N	Smooth ball Z-force slightly strongest

Torque Y	3.95×10^{-5}	5.63×10^{-5}	$2.82 \times 10^{-5} \text{ N}\cdot\text{m}$	It. 1 highest torque at slow speed
Resultant Force	~0.456 N	~0.198 N	~0.483 N	It. 2 dominant; It. 1 weakest
Wake Character	Organized helical	Fragmented vortices	Coherent spiral	It. 2 most controlled

Table 13. Regular throw condition comparison across all three iterations.

At regular throw conditions, Iteration 2's advantage is even more pronounced. It produces the highest resultant force of any design, including the smooth ball, with the largest lateral Y-force and a Z-force close to the smooth ball baseline. Iteration 1 is the weakest performer at slow speed, generating less than half the resultant force of either competitor.

8. Aerodynamic Interpretation

8.1 The Role of the Magnus Effect

The smooth ball baseline makes clear that spin-induced aerodynamic force is the dominant mechanism at work across all designs. A spinning sphere at 2000 RPM and 30 m/s already produces approximately 16.7 N of total aerodynamic force, almost entirely in the Z-direction. The vanned designs are therefore best understood as modifiers of this baseline Magnus response.

8.2 Iteration 1: Turbulence-Driven Lateral Asymmetry

The through-holes create localized flow separation and reattachment zones that interact with vane-induced rotation to produce strong lateral Y-forces. The holes allow internal pressure equalization at specific surface locations, creating asymmetric stagnation and wake zones that generate side force. However, this also disrupts the clean Magnus Z-force, and the highly turbulent fragmented wake may produce less predictable trajectory behavior.

8.3 Iteration 2: Vane-Guided Directional Steering

The thin, high-count vanes function more like an impeller, guiding airflow along defined spiral paths as the ball rotates. This produces a more coherent pressure gradient and a well-organized trailing vortex system. The Magnus Z-force is largely preserved while meaningful lateral force is added — a combination expected to produce more predictable and controllable trajectory break.

8.4 Design Recommendation (Pre-Physical Testing)

Based on the CFD results, Iteration 2 is the more promising design for a controlled, predictable trajectory break across both throwing speeds. It preserves aerodynamic efficiency at slow speed (where Iteration 1 largely fails), while adding meaningful multi-directional force at fastball speed without sacrificing the Magnus-driven Z-force. Iteration 1 may be of interest if a specifically strong lateral break at high speed is the design goal.

9. Reynolds Number Analysis and Flow Regime

9.1 Reynolds Number Definition and Calculation

The Reynolds number (Re) is the fundamental dimensionless parameter governing whether a flow is laminar or turbulent. For a sphere in external flow it is defined as:

$$Re = (\rho \cdot U \cdot D) / \mu$$

Where ρ is the air density (1.225 kg/m³ at standard sea-level conditions), U is the freestream velocity, D is the ball diameter (0.072 m), and μ is the dynamic viscosity of air (1.81 × 10⁻⁵ Pa·s). Applying these values to both simulation conditions:

$$\text{Sim 1 (30 m/s): } Re = (1.225 \times 30 \times 0.072) / 1.81 \times 10^{-5} \approx 146,000$$

$$\text{Sim 2 (17.88 m/s): } Re = (1.225 \times 17.88 \times 0.072) / 1.81 \times 10^{-5} \approx 87,000$$

Both values fall well within the subcritical turbulent regime for bluff-body aerodynamics, which for a sphere spans roughly $Re = 10^4$ to 4×10^5 . At $Re \approx 146,000$, the boundary layer remains laminar over much of the windward hemisphere before separating into a turbulent wake — precisely the regime in which vane and hole geometry have maximum influence on separation location, wake width, and asymmetric force generation. The “drag crisis” associated with a smooth sphere does not occur until $Re \approx 4 \times 10^5$ (approximately 82 m/s for this ball diameter), confirming that neither simulation condition reaches the supercritical flow regime.

9.2 Turbulence Model and Flow Regime Implications

SOLIDWORKS Flow Simulation applies the k- ϵ (k-epsilon) turbulence model with automatic wall treatment. At the Reynolds numbers present in this study — $Re \approx 87,000$ – $146,000$ — the flow around the ball is fully turbulent in the separated wake region. The boundary layer over the windward hemisphere transitions from laminar to turbulent somewhere on the rear hemisphere depending on local surface geometry, and the k- ϵ model with its two transport equations (turbulent kinetic energy k and turbulent dissipation rate ϵ) is an appropriate choice for capturing the time-averaged behavior of such wakes.

The key implication of fully turbulent wake flow for this study is that vane geometry does not need to trigger turbulence — it already exists at these Reynolds numbers. Instead, the vanes function by redirecting turbulent momentum within the boundary layer and near wake, biasing the wake asymmetry and thus the pressure distribution around the ball’s surface. Iteration 1’s through-holes act as additional turbulence generators, promoting early separation at specific surface locations and creating strongly asymmetric stagnation zones. Iteration 2’s thin vanes, by contrast, primarily redirect flow direction rather than promote additional separation, producing a more organized turbulent wake with higher net directional force.

The spin parameter ($S = R\omega / U$, where R is ball radius and ω is angular velocity in rad/s) quantifies the relative contribution of spin to flight aerodynamics. For Sim 1: $S = (0.036 \times 209.4) / 30 \approx 0.25$. For Sim 2: $S = (0.036 \times 52.4) / 17.88 \approx 0.11$. Values of S above 0.1 are generally sufficient to produce a measurable Magnus force on a smooth sphere. The vanned designs augment this Magnus baseline by coupling surface-induced flow asymmetry with the underlying spin-driven pressure differential.

10. Mesh Configuration and Sensitivity

10.1 Mesh Setup

SOLIDWORKS Flow Simulation uses a Cartesian immersed boundary mesh with automatic local refinement at solid-fluid interfaces. For all simulations in this study, a global mesh level of 5 was applied, with local refinement to level 7 at the ball surface and all vane and hole features. This produces a minimum cell size of approximately 0.44 mm near vane edges — sufficient to resolve the primary vortex structures shed from vane trailing edges and hole boundaries, though insufficient to capture the smallest turbulent scales (Kolmogorov scale), which are handled by the turbulence model closure rather than by direct resolution.

The computational domain was set to a rectangular external flow volume extending approximately 5 ball diameters upstream, 10 diameters downstream, and 5 diameters laterally in all directions. These dimensions are consistent with best practices for external sphere aerodynamics in SOLIDWORKS Flow Simulation and ensure that inlet and outlet boundary conditions do not artificially constrain the developing wake. The rotating reference frame was applied via global rotation to simulate spin, consistent with the steady-state solver framework.

10.2 Mesh Sensitivity Discussion

A formal mesh sensitivity (grid convergence) study was not conducted for this preliminary phase, as the primary objective is comparative geometry evaluation rather than absolute aerodynamic quantification. However, several qualitative indicators support the adequacy of the mesh for this purpose. First, the smooth ball baseline results are consistent with expected Magnus force magnitudes for a 72 mm sphere at the given velocity and spin conditions, confirming the mesh is not grossly under-resolved. Second, convergence quality was observably better in geometrically simpler configurations (smooth ball, Iteration 2) than in Iteration 1, which is expected: complex vortex shedding from the hole-and-vane geometry produces unsteady flow structures that a steady-state solver with finite mesh resolution cannot fully capture, resulting in oscillating residuals and lower goal convergence percentages.

For future CFD phases, a structured mesh sensitivity study is recommended: running each geometry at two or three successively refined mesh levels (e.g., global levels 5, 6, and 7) and comparing resultant force values until the percentage change between refinement levels falls below 5%. This would provide a quantitative basis for selecting the minimum mesh level that achieves grid-independent results, and would allow force values to be reported with explicit numerical uncertainty estimates. A transient simulation approach for Iteration 1 at fastball conditions would also be valuable, as the steady-state solver is fundamentally ill-suited to capturing periodic vortex shedding — the dominant flow feature of that design at high speed.

11. Consolidated Aerodynamic Force Summary

The table below consolidates all resultant aerodynamic forces across every design iteration and both simulation conditions into a single reference. Resultant force magnitudes are computed as the vector sum of the X, Y, and Z force components: $F_v = \sqrt{F_x^2 + F_y^2 + F_z^2}$. All values are drawn directly from the SOLIDWORKS goal summaries reported in Sections 4 through 6.

Iter. 1 — Sim 1 (Fastball)	-2.32	-9.13	-11.30	14.71	17–100%
Iter. 1 — Sim 2 (Regular Throw)	+0.070	-0.106	-0.152	0.198	14–26%
Iter. 2 — Sim 1 (Fastball)	-5.03	-5.28	-15.21	16.89	30–95%
Iter. 2 — Sim 2 (Regular Throw)	-0.279	-0.283	-0.275	0.483	30–83%
Smooth Ball — Sim 1 (Fastball)	-5.08	-0.21	-15.87	16.66	High
Smooth Ball — Sim 2 (Regular Throw)	-0.320	-0.0159	-0.325	0.456	High

Table 14. Consolidated resultant aerodynamic force summary. Green shading = highest resultant in that speed category. Yellow = second. Resultant = $\sqrt{F_x^2 + F_y^2 + F_z^2}$.

The consolidated table makes the velocity dependence immediately visible: fastball resultant forces are 35–74 times larger than regular-throw resultant forces across all designs. At fastball conditions, Iteration 2 produces the highest total resultant force at 16.89 N, marginally exceeding the smooth ball (16.66 N) and substantially exceeding Iteration 1 (14.71 N). At regular throw conditions the ordering is preserved: Iteration 2 leads at 0.483 N, followed by the smooth ball at 0.456 N and Iteration 1 at a distant 0.198 N. Iteration 1’s strong fastball lateral force (–9.13 N in Y) is notable but comes at the cost of reduced resultant force overall and near-complete aerodynamic inefficiency at lower speeds.

12. Practical Applications and Engineering Relevance

Although this study was motivated by a recreational design question, the aerodynamic principles it investigates have direct relevance across several applied engineering domains. The Wendiball geometry explores a design space that sits at the intersection of sports engineering, flow-control research, and projectile aerodynamics.

12.1 Sports Engineering and Pitch Development

The most direct application of this work is in the design of training and recreational pitching balls. The quantified relationship between surface geometry and trajectory break angle could inform the development of balls specifically engineered to simulate particular pitch types — a curveball-replica ball, for example, or a ball designed to produce a sweeper-like lateral break — for use in batting practice or pitching mechanics training. Unlike a standard wiffleball, where hole placement is not optimized for any specific trajectory, a purpose-designed vaned ball could produce a consistent and repeatable break on every throw, independent of subtle grip variations. This would give coaches and players a controlled, reproducible tool for developing pitch recognition at the plate.

12.2 Controllable Recreational Projectiles

More broadly, the Wendiball concept represents a class of projectile whose trajectory is engineered through geometry rather than controlled entirely by the thrower’s grip and release. Surface vane patterns and hole arrays could be designed to produce specific break magnitudes and directions for different recreational games, target sports, or backyard skill challenges. A family of ball designs with progressively larger breaks — from subtle to extreme — could serve as a skill development ladder

for pitch control. The CFD results from this study provide an initial quantitative foundation for such a design space: Iteration 2 produces a controlled multi-directional break, while Iteration 1 produces a dominant lateral break at high speed. Future iterations exploring the parameter space between these two geometries could map out a full range of break characters achievable with vane and hole geometry alone.

12.3 Aerodynamic Flow-Control Research

The surface geometry concepts explored in this project — specifically passive vane arrays that redirect boundary layer flow without external energy input — are directly analogous to passive flow-control devices used in aerospace and industrial applications. Vortex generators on aircraft wings, trip strips on wind turbine blades, and dimples on golf balls all exploit the same principle: small geometric features that deliberately manipulate boundary layer behavior to shift separation location, reduce wake width, or bias lift and drag forces. The Wendiball vane geometry, at a larger and more aggressive scale, is effectively an extreme version of this class of device. Results from this study — particularly the vorticity isosurface data and the comparison of coherent versus fragmented wake structures between iterations — could inform the design of passive flow-control geometries in applications where a rotating or spinning body is exposed to a freestream: wind energy capture devices, spinning sensor housings, or aerodynamically stabilized payloads.

12.4 Pitching Aids and Instructional Technology

A geometry-encoded trajectory break has a practical advantage over grip-dependent break: it removes a variable. For pitching instruction specifically, a ball that produces a known break regardless of grip takes grip execution out of the equation and allows coaches to isolate the mechanical aspects of the throwing motion. This could be particularly useful at youth development levels, where grip consistency is difficult to establish and where understanding the feel of a break-pitch trajectory is more valuable than learning a specific finger placement. Beyond instruction, a controlled-break ball with a known aerodynamic profile could serve as a standardized test object for pitching machine calibration, spin-rate measurement system validation, or catcher reaction-time research.

13. Key Findings

13.1 Velocity and Spin Sensitivity

All vaned designs are strongly sensitive to throw velocity and spin rate. The fastball condition produces resultant forces approximately 35–74 times larger than the regular throw condition, depending on geometry. This reflects the quadratic velocity dependence of aerodynamic forces compounded by increased spin rate effects.

13.2 Geometry Effects

- Through-holes (Iteration 1): Generate lateral Y-force through localized boundary layer disruption. Highly effective at fastball speed, near-negligible at regular throw speed. Wake is chaotic and turbulent.
- Thin high-count vanes (Iteration 2): Guide flow coherently, preserving Z-force while adding lateral force. Effective across both speed regimes. Wake is organized and directional.

- Smooth ball (Iteration 3): Establishes the Magnus baseline. All design modifications are measured against this reference.

13.3 Convergence Observations

Convergence varied significantly across simulations. Fastball cases generally showed more variable convergence due to strong vortex shedding regimes which are inherently difficult for steady-state solvers. The smooth ball and Iteration 2 at regular throw showed the most stable convergence. All convergence behavior is noted in the simulation goal tables and should be considered when interpreting quantitative force values.

14. Next Steps and Future Work

10.1 Physical Prototype Testing (Phase 2)

- Trajectory recording using high-speed video or motion capture to measure actual break angle and magnitude at both throwing speeds.
- Force plate or instrumented net measurements to quantify total aerodynamic deflection force.
- Spin-rate calibration using a consistent throwing mechanism to ensure RPM inputs match simulated conditions.
- Mass-normalized performance comparison, accounting for the ~1.35 g mass difference between iterations.

14.2 Potential CFD Refinements

- Local mesh refinement around vane edges and hole boundaries to improve convergence in high-gradient regions.
- Transient simulation for fastball cases to capture vortex shedding dynamics that the steady-state solver cannot fully resolve.
- Iteration 4 design exploration: a hybrid combining Iteration 2's thin vane count with a small number of strategically placed holes to introduce targeted lateral asymmetry without the wake fragmentation of Iteration 1.

15. Conclusion

This report presents a preliminary CFD study of the Wendiball, an original aerodynamic ball design. Three design iterations and a smooth-ball baseline were evaluated across two throwing conditions using steady-state SOLIDWORKS Flow Simulation. The results demonstrate that surface geometry has a significant and measurable impact on aerodynamic force distribution, with design-dependent effects that vary substantially with throw velocity and spin rate.

Iteration 2 — the thin, high-count spiral vane design without through-holes — emerges as the strongest candidate for physical prototype testing. It produces the highest total aerodynamic force across both simulation conditions, preserves the Magnus-driven directional Z-force, and adds meaningful lateral force through organized vane-guided flow.

Iteration 1 — the spiral vane design with through-holes — produces the strongest lateral Y-force at fastball conditions but is significantly less effective at regular throw speed and generates a chaotic turbulent wake. It may be of interest for specialized high-speed applications or as a basis for further hole-geometry optimization.

All results are preliminary and subject to the convergence limitations described in Section 3. Physical experimental validation is the critical next step to confirm and refine these findings.

Wendiball CFD Study — Preliminary Design Report — Wendell Abara — May 2026

Conducted in SOLIDWORKS Flow Simulation. No physical experimental validation has been performed at time of writing.